MATH 152 **EXAM III** NAME _____

Mrs. Bonny Tighe PART ONE

11.8-10.2 TA_____Wed. 5/1/2024

50 points

1.(4 points each) Circle the letter for the correct answer

a) The sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n (\pi)^{2n+1}}{36^n (2n+1)!}$

A) $1/2$

B) $\sqrt{3}/2$

C) **1**

D) $\sin(1/2)$

E) Some other answer

b) The interval of convergence for the series $\sum_{n=0}^{\infty} \frac{(x-1)^n}{n+2}$

A) $(0,2)$

B) $(0,2]$

C) $[0,2]$

D) $[0,2)$

E) Some other answer

c) The radius of convergence for the series $\sum_{n=1}^{\infty} \frac{4^n x^n}{5^{n+1}}$

A) $4/5$

B) $(-\infty, \infty)$

C) $5/4$

D) $\{0\}$

E) Some other answer

d) Which of the following parabola has its vertex at (0, 0) and the directrix of $x = 2$

A) $x^2 = 8y$

B) $x^2 = -8y$

C) $y^2 = 8x$

D) $y^2 = -8x$
E) Some other answer

e) The second derivative of the parametric equation $x = 2 - t^2,\ y = 3 + 3t$ is:

A) $6/t^4$
B) $-3/t^4$
C) $t^3/6$
D) $6/t^3$
E) Some other answer

f) The derivative of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n)!}$ is

A) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n-1}}{(2n)!}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n}}{(2n)!}$
C) $\sum_{n=1}^{\infty} \frac{(-1)^n (2n+1) x^{2n-1}}{(2n)!}$
D) $\sum_{n=1}^{\infty} \frac{(-1)^n (2n+1) x^{2n}}{(2n)!}$
E) Some other answer

g) The Maclaurin series representation for $f(x) = \cos(3x^2)$ is:
A) $\sum_{n=0}^{\infty} \frac{(-1)^n 3x^{4n+2}}{2n}$
B) $\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n} x^{2n+1}}{(2n)!}$
C) $\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n} x^{4n+2}}{(2n+1)!}$

D) $\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n} x^{4n}}{(2n)!}$

E) Some other answer

h) The integral of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n}$ is

A) $C + \sum_{n=0}^{\infty} \frac{(-1)^n (2n+2) x^{2n+2}}{2n}$

B) $C + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+2}}{(2n)(2n+2)}$

C) $C + \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n)}$

D) $C + \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{(2n)(2n)}$

E) Some other answer

i) Which of the following ellipse has its foci at (-1, -1) and (-5, -1) and vertex of (1,-1)

A) $\frac{(x+3)^2}{12} + \frac{(y+1)^2}{16} = 1$

B) $\frac{(x+3)^2}{16} + \frac{(y+1)^2}{12} = 1$

C) $\frac{(x-3)^2}{16} + \frac{(y-1)^2}{12} = 1$

D) $\frac{(x-1)^2}{4} + \frac{(y-1)^2}{1} = 1$

E) Some other answer

2. (2 points each) True or False

a) In the Comparison test, if $0 \le f(x) \le g(x)$ and $g(x)$ is convergent then we conclude that $f(x)$ is also convergent. _____

b) The Maclaurin series for $e^x = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}$ _____

c) The asymptotes for the hyperbola $\frac{(y-1)^2}{16} - \frac{(x+2)^2}{9} = 1$ are $y = \pm\frac{4}{3}x$ _____

d) Given $f(2) = 1$, $f'(2) = 2/3$, $f''(2) = -25/3^2$, $f'''(2) = 258/3^3$, etc.

then $f^{(n)}(2) = \frac{(-1)^n (2 \cdot 5 \cdot 8 \cdots (3n-1)}{3^n}$ _____

e)

f) If $-2 < x < 4$ then $|x-2| < 2$ _____

3. (7 points) Use the Binomial Theorem to find the function as a power series.
$f(x) = \sqrt[3]{1-x^2}$

MATH 152 **EXAM III** NAME _____
Mrs. Bonny Tighe PART TWO
11.4-10.5 TA_____ Wed.5/2/2024
50 points

1.(10 points) Find a Taylor polynomial of degree 3 for $f(x) = \frac{1}{\sqrt{x}}$ centered at $a = 3$.

Use it to approximate $\frac{1}{\sqrt{3.1}}$.

How accurate is this approximation when $1 < x < 5$?

If $\left|f^{(n+1)}(x)\right| \le M$ for $|x - a| \le d$, then the remainder $R_n(x) \le \frac{M}{(n+1)!}\left|x - a\right|^{n+1}$ for $|x - a| \le d$

2. (4 points) Find the center, foci, and vertices. Sketch the graph
$16x^2 + 64x + 4y^2 - 24y + 36 = 0$

3. (4 points) Find the vertices, foci, asymptotes, and center. Sketch the graph.

$4(x+2)^2 - 9(y-1)^2 = 36$

4. (5 points) Use series to evaluate the limit $\lim_{x \to 0} \frac{\cos(x^3) - 1}{x^2 - \arctan(x^2)}$

5. (10 points) a) Sketch the plane curve defined by the given parametric equation. Eliminate the parameter to find a Cartesian equation of the curve. Indicate with an arrow the direction in which the curve is traced as t increases. $x = 3\sin\theta, \quad y = 4\cos\theta$

b) Find the area under the curve on the interval $0 \le \theta \le \pi/2$

6. (4 points) Find the Interval and Radius of convergence for the power series

$$\sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^n}{n+3}$$

7. (4 points) a) Evaluate the indefinite integral as a power series. $\int \frac{x^2}{5+x^2}\,dx$

8. (10 points) Find the Taylor series for $f(x) = \frac{1}{(3+x)^2}$ centered at a = 3

MATH 152 **EXAM IIIA** NAME _____

Mrs. Bonny Tighe PART ONE

11.8-10.2 TA_____ Wed. 5/1/2024

50 points

1. (4 points each) Circle the letter for the correct answer

a) The Maclaurin series representation for $f(x) = x\arctan(x^3)$ is:

A) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+2}}{(2n+1)!}$

B) $\sum_{n=0}^{\infty} \frac{x^{6n+4}}{2n+1}$

C) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+4}}{(2n+1)!}$

D) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+4}}{2n+1}$

E) Some other answer

b) The sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n (\pi)^{2n}}{3^{2n}(2n)!}$

A) $\pi/3$

B) $3\sqrt{3}/2$

C) $\sqrt{3}/2$

D) $1/2$

E) Some other answer

c) The radius of convergence for the series $\sum_{n=1}^{\infty} \frac{2^n x^n}{5^{n+1}}$

A) $2/5$

B) $(-\infty, \infty)$

C) $5/2$

D) $\{5\}$

E) Some other answer

d) Which of the following parabola has its vertex at (0, 0) and the directrix of $y = 1$

A) $x^2 = 4y$

B) $x^2 = -4y$

C) $y^2 = 4x$
D) $y^2 = -4x$
E) Some other answer

e) The second derivative of the parametric equation $x = t^2 + 1, \quad y = 1 - t^3$

A) $-3t/2$

B) $3t/2$

C) $-2/3t$

D) $2t/3$

E) Some other answer

f) The derivative of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$ is

A) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n}}{(2n)!}$

C) $\sum_{n=1}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

D) $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n)!}$

E) Some other answer

2. (2 points each) True or False

a)If the parametric curve $x = f(t)$, $y = g(t)$, satisfies $f'(1) = 0$, then it has a horizontal tangent when $t = 1$. _____

b) The Maclaurin series for $\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2n}$ _____

c) The asymptotes for the hyperbola $\frac{(x-1)^2}{4} - \frac{(y+2)^2}{9} = 1$ are $y = \pm\frac{3}{2}x$ _____

d) Given $f(2) = 1$, $f'(2) = 2/3$, $f''(2) = -25/3^2$, $f'''(2) = 258/3^3$, etc.

then $f^{(n)}(2) = \frac{(-1)^n(2 \cdot 5 \cdot 8 \cdots (3n-1)}{3^n}$ _____

e) If $-2 < x < 5$ then $|x-2| < 3$ _____

3. (8 points) Use the Binomial Theorem to find the function as a power series.

$f(x) = \frac{1}{(1-x^3)^2}$

4. (5 points) a) Find a Taylor polynomial of degree 3 for $f(x) = \sin x$ centered at $a = \pi/6$.

b) (4 points) Use it to approximate $\sin\left(31\pi/180\right)$.

c) (4 points) How accurate is this approximation when $\pi/9 < x < 2\pi/9$?

If $\left|f^{(n+1)}(x)\right| \le M$ for $|x - a| \le d$, then the remainder $R_n(x) \le \frac{M}{(n+1)!}|x - a|^{n+1}$ for $|x - a| \le d$

MATH 152 **EXAM IIIA** NAME _____

Mrs. Bonny Tighe PART TWO

11.8-10.2 TA_____ Wed.5/2/2018

50 points

1. (3 points) Find the vertex, focus and directrix. Sketch the graph.

$(y+1)^2 = -2(x+3)$

2. (4 points) Find the center, foci, and vertices. Sketch the graph
$9x^2 - 18x + 4y^2 + 8y = 23$

3. (4 points) Find the vertices, foci, asymptotes, and center. Sketch the graph.
$4(x+2)^2 - 25(y-1)^2 = 100$

4. (4 points) a) Sketch the plane curve defined by the given parametric equation. Eliminate the parameter to find a Cartesian equation of the curve. Indicate with an arrow the direction in which the curve is traced as t increases. $x = 3\cos\theta, \quad y = 2\sin\theta$

b) (5 points) Find the area under the curve on the interval $0 \le \theta \le \pi/2$

5. (5 points) Find the Interval and Radius of convergence for the power series

$$\sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^n}{n+2}$$

6. (5 points) Use series to evaluate the limit $\lim_{x \to 0} \frac{1 + x - \cos(2x)}{e^x - 1}$

7. (10 points) Find the Taylor series for $f(x) = \ln(x - 2)$ centered at a = 4

8. (5 points) Evaluate the integral as a power series and approximate to T_5 .

$$\int_0^{0.5} \frac{x^2}{3 - x}\, dx$$

9. (10 points) Compute the surface area obtained by rotating the given curve about the **y-axis** $x = e^{t} + e^{-t}, \quad y = 5 - 2t, \quad 1 \le t \le 0$

$$S = 2\pi \int_{\alpha}^{\beta} radius \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$